

Roll No.

Total Pages : 3

002606

May, 2023

B.Tech. (Civil) VIth Semester

Air Noise Pollution Control (PEC-CV-403-1)

Time : 3 Hours]

[Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part ~B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. ☒ (a) What are the global implications of air pollution? (1.5)
- ☒ (b) List the classification of air pollutants based on their types. (1.5)
- ☒ (c) Mention two methods used for the control of particulate air pollutants. (1.5)
- (d) What are the basics of acoustics and how are sound intensity and sound pressure levels measured? (1.5)
- ☒ (e) What are the common gaseous contaminants and their sources in air pollution? (1.5)

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- (f) Explain the concept of psychoacoustics and its relevance in noise pollution. (1.5)
- ✓(g) Define air pollution dispersion. (1.5)
- ✓(h) Explain Wet collectors. (1.5)
- (i) Define Acoustic. (1.5)
- ✓(j) Write effect of Noise pollution on human health. (1.5)

PART-B

- 2. ✓(a) List and explain the types of Plume Behaviours. (10)
- ✓(b) Explain indoor air & noise pollution. (5)
- 3. (a) Explain Lapse Rate and its types using numerical expressions. (5)
- (b) Derive the expression for stack height in Gaussian Plume Model. (10)
- ✓4. Explain the composition and structure of atmosphere. (15)
- 5. (a) What do you mean by Air sampling. How can it be measured? (5)
- (b) List and explain (using diagrams) any two control devices for particulates. (10)

- 6. ✓(a) A noise meter located at a distance of 30 m from a point source recorded 74 dB Find the reading at a distance of 60m from the point source. (10)
- ✓(b) List and explain the sound characteristics. (5)

- ✓7. Write in detail about control of gaseous contaminants. (15)
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